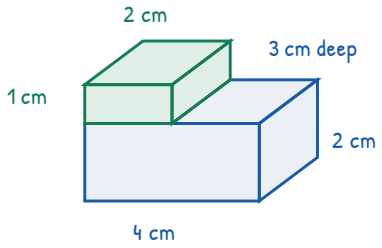
Add volumes of nonoverlapping prisms

I can… decompose a solid into two prisms and add their volumes.

Split a step-shaped solid

A lower prism is 4 cm × 3 cm × 2 cm. A top prism is 2 cm × 3 cm × 1 cm. The top sits on the lower prism.



Blue: lower prism · Green: upper prism

Find each part

Lower: $4 \times 3 \times 2 = 24$ cm³

Top: $2 \times 3 \times 1 = 6$ cm³

Add without double counting

$24 + 6 = 30$ cm³

The prisms touch but their interiors do not overlap.